Physics-guided characterisation-time budgeting for second-life LFP cells: how few cycles does PyBaMM's reaction-limited SEI need to predict remaining life?

Krishna Chaitanya Vaddepally
Turno (Blubble Pvt. Ltd.), India

Second-life repurposing of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost: pack builders must estimate remaining useful life (RUL) with no beginning-of-life data and limited cycling time. Prior PyBaMM-based work has optimised calibration [3] and generalised physics-based degradation models [4] but has not quantified the cycling budget required for reliable RUL prediction. We ask: *how many measured cycles does a physics-based degradation model need before it can predict the remaining life of a used cell?*

We calibrate PyBaMM's [1] reaction-limited SEI degradation model (`Prada2013` chemistry [2], isothermal 25 °C) against the measured fade slope of a used cell using only its first K cycles, then run the calibrated model to the full measured horizon and score against the held-out cycles K–N. A single scalar (SEI exchange current density j_{SEI}) is fit per cell by bisection on $\log_{10}(j_{SEI})$. Per-cell open-circuit voltage, initial state-of-health, and ohmic resistance are constrained from HPPC and OCV–SoC characterisation. We sweep $K \in \{50, 100, 200, 400, 800\}$ across seven MFR_A-cohort cells with clean long-trajectory records ($N > 1000$ cycles each) that pass an upstream reaction-limited-shape filter. Calibration and hold-out sweeps are orchestrated by a Claude Code multi-agent pipeline (`AGENT_PARAM_ID`, `AGENT_VOLTARIS_TUNING`, `AGENT_SIMULATION`) released with the code.

Prediction RMSE on held-out cycles drops from a median of 11.2 pp SoH at $K = 50$ to 1.4 pp at $K = 100$ and 0.5 pp at $K = 800$ (Fig. 1). At $K = 400$, all 7/7 cells achieve $RMSE_{test} < 3$ pp SoH: **400 vs. ≈ 1200 measured cycles per cell (3$\times$ reduction, ≈ 4 months of test-lab time saved at 0.25 C).** Five of seven cells already converge at $K = 100$ with end-of-trajectory error below 0.4 pp (Fig. 2), yielding a **12$\times$ reduction** for that sub-cohort. The remaining two exhibit a post-formation recovery transient (positive apparent slope across cycles 50–200) and require $K \geq 400$; this signal itself classifies cells for correct calibration windowing.

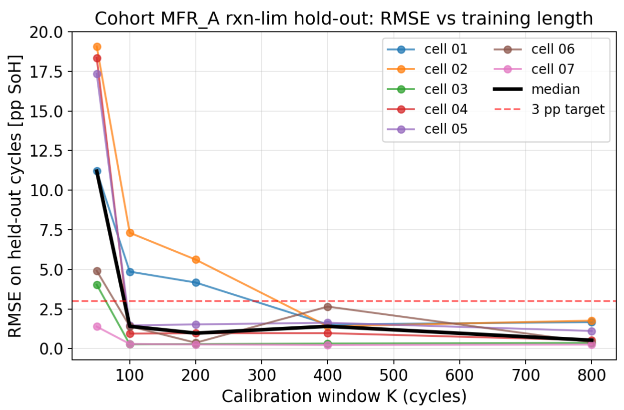

Figure 1: Held-out RMSE vs. training window K; red dashed = 3 pp SoH target.

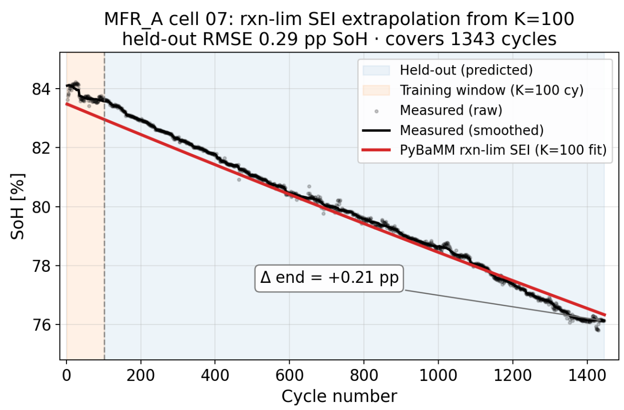

Figure 2: K=100-cycle fit predicts remaining 1345 cycles of cell 07 to 0.29 pp SoH RMSE.

The result holds only within the reaction-limited SEI regime. Cells flagged by an upstream shape filter (non-monotonic-early, three-regime, or accelerating anomalies) are excluded — such shapes may reflect LAM domination (where PyBaMM's stress-driven LAM lacks saturation) or data-processing artefacts (batch-transition discontinuities). The full 35-run sweep is reproducible on a single core in ≈ 30 min using PyBaMM `v26.4.3`.

References. [1] V. Sulzer et al., *Python Battery Mathematical Modelling (PyBaMM)*, J. Open Res. Softw. 9(1), 14–21 (2021). [2] M. Prada et al., *Simplified electrochemical and thermal model of LiFePO$_4$-graphite Li-ion batteries*, J. Electrochem. Soc. 160(4), A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Lithium-Ion Batteries*, PyBaMM Conf. 2025 Book of Abstracts, p. 19. [4] J. A. Kuzhiyil et al., *Enhancing Generalisability of Physics-based Battery Degradation Using UDE*, PyBaMM Conf. 2025 Book of Abstracts, p. 12.